

UI/UX for Planetarium & GNSS

Competitor Research & Analysis Report

Prepared by: Aleeza Noor | Student ID: 242802

Date: July 7, 2026

Table of Contents

1. Introduction
2. Objectives
3. Competitor 1 – Shanghai Astronomy Museum
 - 3.1 Home Page
 - 3.2 Booking
 - 3.3 Navigation
 - 3.4 Design Decisions
4. Competitor 2 – Adler Planetarium
 - 4.1 Home Page
 - 4.2 Interactive Museum Map
 - 4.3 Calendar
 - 4.4 Museum Programs
 - 4.5 Dome Shows
 - 4.6 Venue Rentals
 - 4.7 Research
 - 4.8 Space Shop
 - 4.9 Icon Legend
 - 4.10 Design Decisions
5. Comparative Analysis
6. Overall Findings
7. Features to Adopt
8. Features to Avoid
9. Conclusion
10. References

1. Introduction

This document forms part of the competitor research phase for the “UI/UX for Planetarium & GNSS” project. As outlined in the project brief, this phase involves studying existing planetarium and observatory websites to understand common design patterns, navigation structures, and feature sets before beginning wireframing in Figma. This comprehensive report synthesizes structural, navigational, visual, and transactional data from premier global astronomical institutions to inform the design and technical decisions of the upcoming Planetarium & Observatory Management System.

2. Objectives

- Identify industry-standard and unique UI/UX patterns across premier astronomical and observatory platforms.
- Analyze user onboarding, navigation frameworks, and transactional conversion funnels (ticketing, reservation, and donations).
- Evaluate microinteractions and spatial accessibility features that enhance user engagement and trust.
- Establish a definitive framework of design features to adopt or avoid during the system's wireframing and prototyping phases.

3. Competitor 1 – Shanghai Astronomy Museum

The Shanghai Astronomy Museum (SAM) is the largest astronomy museum in the world. The website functions primarily as an informational and promotional platform rather than a transactional one. Its core role is limited to information and marketing, while its transactional layers are systematically offloaded to external platforms.

3.1 Home Page

- **First Impression:** The homepage opens with a full-screen cinematic hero: a dark night sky, scattered stars, and an illustrated render of the museum building's signature dome/sphere structure. It features a bilingual title and a subtle scroll-down indicator. There is no immediate “Book Now” call-to-action competing for attention, intentionally prioritizing atmosphere and architecture over transactional prompts.

Structure & Hierarchy

1. **Hero:** Building illustration, bilingual title, scroll indicator, and randomly appearing “shooting star” animations.
 2. **Three-Card Panel:** News Announcement, Latest Activities, and Latest Courses (icon and text label overlaid on an image background).
 3. **Three-Column Feature Block:** Temporary Exhibition, Permanent Exhibition, and Cinema Information.
 4. **Interactive Constellation Viewer:** A stargazer-silhouette scene paired with a numbered, paginated zodiac carousel.
 5. **Interactive Moon-Phase Calendar Widget:** A lunar calendar providing sunrise, sunset, moonrise, and moonset times. Tapping a specific date triggers a tooltip popup displaying the exact moon phase and time.
 6. **Footer:** Service information (address, phone, hours), an expansive sitemap of link columns (Visit, Exhibition, Exploration), decorative astronaut-on-the-moon imagery, and copyright/registration details.
- **Visual & Typography Patterns:** The homepage employs a deep navy and black base with orange/gold as the primary accent color for hover states, active tab underlines, and icon dots. Blue serves as a secondary accent color for buttons and hover states on feature cards. Section headings are paired with a small circular orange “sun” icon placed beside the heading text (e.g., “● Welcome Address”), establishing a consistent layout signature. Body copy is left-aligned, sans-serif, and styled in short paragraphs to ensure optimal readability.

3.2 Booking

- **Transactional Infrastructure:** Notably, no ticket purchases, cinema bookings, or reservations are completed on the website itself. Every “Buy Ticket,” “Ticket,” and “Reservation” action triggers a modal overlay displaying a WeChat mini-program QR code instead of an in-browser checkout pipeline.
- **UX Fractures:** The web application separates transactional actions visually from informational paths, which aids discovery. However, both “Ticket” and “Reservation” buttons execute the exact same WeChat QR code modal. This duplication introduces user confusion, as it incorrectly implies distinct workflows for booking vs. ticketing. Additionally, complete dependence on the external WeChat ecosystem limits usability for international or domestic visitors who lack active WeChat profiles.

3.3 Navigation

- **Primary Framework:** Persistent top navigation bar containing: Home, Visit, Exhibition, Exploration, Star Friends Circle, and Cultural and Creative Industries.
- **Utility Infrastructure:** A language toggle (EN/CN), search icon, and account icon reside independently in the top-right terminal of the header interface.
- **Actionable Isolation:** A dedicated “Ticket | Reservation” button is visually boxed and isolated in blue within the primary navigation array. This keeps the site's primary conversion target visible at all times, preventing it from getting buried during deep scroll cycles.
- **Secondary Layers:** The Exhibition segment deploys an expansive mega-menu dropdown containing Permanent Exhibition, Temporary Exhibition, Characteristic Exhibition, Cinema, Collection, and Virtual Exhibition Hall pathways. Deep-page navigation utilizes sticky sub-navigation tab bars (e.g., Overview/Service/Visit Plan; Current/Upcoming/Past) that pin to the viewport during active scrolling to allow seamless cross-page transitions.

3.4 Design Decisions

The SAM architecture shifts its visual theme across distinct sections, utilizing three separate color moods:

- **Dark-Cosmic Mood (Home):** Deep navy/black with orange accents and high-resolution space/nebula photography.
- **Purple-Utility Mood (Visit):** Purple and lavender dominate the tab bars and backgrounds, using white content cards and real architectural wayfinding photography.
- **Clean-White Mood (Exhibition/Collections):** Minimalist white/light-gray backgrounds with product-style specimen photography shot on plain white backdrops.

Microinteractions & Animation Patterns

- **Atmospheric Flourishes:** Randomly timed “shooting stars” animate across the homepage hero to visually support the cosmic theme.
- **Hover States:** Feature cards highlight and change color to gold/orange upon user hover. In parallel, rotating circular icon animations run inside the cards to guide the user's focus.
- **Progressive Disclosure:** Homepage exhibition panels feature hover-responsive text fades, revealing context and keeping layouts clean until the user acts.

- **Scroll Reveal:** The Permanent Exhibition pages utilize scroll-triggered reveal transformations. Text and image blocks slide upward or downward while fading into the viewport, which adds polish to heavy text layouts.
- **Hover Zoom:** Specimen images within the Collections gallery scale upward slightly within their card bounds on hover. However, this effect remains decorative, as it fails to reveal finer details of the artifacts.

4. Competitor 2 – Adler Planetarium

The Adler Planetarium web platform balances its identity as both a popular public visitor attraction and a serious research and educational institution.

4.1 Home Page

- **First Impression:** The homepage opens with an oversized typography-first hero section displaying the phrase “FIND YOUR SPACE”. The word “SPACE” is highlighted in the brand's signature high-saturation yellow over a background sky photograph. This design establishes an emotional, playful, and confident value proposition before presenting transactional inputs.

Structure & Funnel Hierarchy

1. **Hero Interface:** Value-proposition copy paired with persistent primary Call-to-Actions (CTAs) for Tickets and Donations.
 2. **Task-Oriented Quick Links:** “Visit the Adler” quick links (Plan Your Visit, Dome Shows, Exhibits, Field Trips) sit immediately below the fold to let users self-select their paths quickly.
 3. **Current Programming:** An upcoming events section arranged in a 3-column, date-stamped card grid.
 4. **Cinematic Promotion:** A single full-width feature banner highlighting the primary dome show (“Worlds Beyond Earth”).
 5. **Credibility Content Hub:** A tabbed interface titled “Discover the Sky We All Share,” dividing content into Astronomy History, Adler Research, Astronomy At Home, and Science Activities.
 6. **Monetization Cross-Sell:** A dedicated conversion section promoting Chicago CityPASS.
 7. **Editorial Content:** “More From the Adler” blog and publication matrix.
 8. **Retention Mechanism:** An integrated newsletter signup interface (First Name, Last Name, Email, Submit).
 9. **Footer:** Location data, operating hours, social handles, and an organized sitemap.
- **Visual & UI Infrastructure:** The site uses a high-contrast palette consisting of black, white, and a single saturated yellow accent color. It features two distinct typography families: a bold, condensed sans-serif for institutional headers and a handwritten, marker-style script for humanized sub-captions and taglines. The layout uses full-bleed horizontal sections stacked vertically, with content grouped into distinct rows to prevent cognitive fatigue.

4.2 Interactive Museum Map

- **Structural Layout:** The digital map interface uses an immersive, full-width interactive grid engine designed to help visitors orient themselves within the physical building layout.

- **Component Architecture:** The interface includes floating zoom controls (+/-), floor-level toggle menus, and interactive location pins. Clicking a pin reveals a popover container complete with room descriptions, current exhibit schedules, and direct links to related event detail pages.
- **UX Execution:** The interface helps users transition smoothly from digital trip planning to physical navigation. To keep things clear, the system filters out extraneous data until a user clicks on a specific location pin.

4.3 Calendar

- **Structural Layout:** Built around a multi-view calendar grid that supports monthly, weekly, and scannable list-view configurations.
- **Component Architecture:** Features segmented control bars for view switching, persistent filter dropdown menus (categorized by Event Type, Audience, and Cost), and date-picker modules.
- **UX Execution:** Event instances use card grids with clear date stamps, titles, and quick-booking actions. The interface handles complex, recurring museum events without cluttering the page layout.

4.4 Museum Programs

- **Structural Layout:** Organized as a content library utilizing a 3-column card grid configuration to showcase educational and public program tracks.
- **Component Architecture:** Employs pill-shaped audience tag overlays (e.g., “K-12,” “Adults Only”), responsive card containers that elevate slightly on hover, and clear, descriptive action buttons.
- **UX Execution:** Grouping layouts by target demographic helps target audiences—such as educators, parents, or academic scholars—find relevant programs efficiently.

4.5 Dome Shows

- **Structural Layout:** Designed as a media-rich digital display page utilizing alternating full-width promotional banners and clean 2-column info blocks.
- **Component Architecture:** Integrates embedded HTML5 video preview players, running-time badges, and high-visibility ticket booking buttons.
- **UX Execution:** The page uses cinematic visual elements to drive conversions, putting key details like showtimes and age recommendations right next to the booking triggers.

4.6 Venue Rentals

- **Structural Layout:** A split-screen, photo-forward landing layout designed to market the facility for corporate events, private galas, and weddings.
- **Component Architecture:** Features multi-image galleries with touch-responsive photo carousels, downloadable PDF floor plans, and interactive inquiry forms.
- **UX Execution:** The layout prioritizes visual proof of the space, placing venue capacities and amenity lists close to the primary contact forms.

4.7 Research

- **Structural Layout:** An academic, text-focused portal designed as a 2-column article index with a left-hand sticky filter menu.
- **Component Architecture:** Includes search filters for research areas, clean text publication links, and author profile cards.
- **UX Execution:** Shifting away from playful commercial branding, this section uses clean typography and generous whitespace to support reading and research exploration.

4.8 Space Shop

- **Structural Layout:** An integrated e-commerce layout featuring a 4-column product grid display.
- **Component Architecture:** Includes product filtering sidebars (price, category, availability), product image hover-swaps, shopping cart status displays, and star-rating badges.
- **UX Execution:** The layout follows standard e-commerce patterns, keeping the path from product discovery to checkout clear and familiar.

4.9 Icon Legend

- **Structural Layout:** A clean 2-column reference grid embedded in informational and ticketing footers.
- **Component Architecture:** Maps flat vector icons directly to text definitions, including accessibility markers, facility amenities, and ticket restrictions.
- **UX Execution:** Centralizing system iconography into a reusable key helps users understand visual indicators throughout the platform.

4.10 Design Decisions

- **CTA Hierarchy:** The platform uses a clear, intentional hierarchy for its calls to action. High-intent primary buttons, such as “Tickets,” are rendered as solid, high-contrast yellow fields. In contrast, secondary actions, like “Donate,” use outlined pill layouts to establish clear visual priority.
- **Navigation Logic:** The main navigation header is limited to five clear categories to ensure high scannability. Lower-priority utility links are shifted to an independent sub-header row.
- **Conversion Optimization:** On deep transactional screens, like the donation checkout page, the primary navigation menu is removed entirely. This conversion-focused choice keeps users focused on the checkout form by removing external exit links.

Microinteractions & Animation Patterns

- **Layered Z-Index Scroll Parallax:** As users scroll down the homepage hero, the foreground museum dome photo rises to overlap the lower portion of the “FIND YOUR SPACE” text header, creating an immersive sense of depth.
- **Orbital Path Animations:** Embedded SVGs animate along Bezier curves to depict moons moving in continuous orbits within the dome show section, adding context to space-themed pages.
- **Scroll-Triggered Reveals:** Vector planet illustrations and descriptive content blocks fade into place via intersection observers only when scrolled into the active viewport.

5. Comparative Analysis

The table below contrasts the interface layout and user flows of both competitor platforms:

Evaluation Dimension	Shanghai Astronomy Museum (SAM)	Adler Planetarium
Primary System Role	Informational & Promotional Marketing	Transactional Conversion & Institutional Authority Hub
Visual Architecture	Dynamic multi-mood themes shifting across pages	Single high-contrast brand framework (Black/Yellow)
Navigation Model	Expansive mega-menu with multi-tier sticky sub-tabs	Streamlined primary nav header with top utility row
Checkout Flow	External WeChat mini-program via modal QR codes	Native web checkout paths with localized currency tools
CTA Design	Visual duplication: Multiple labels trigger the same action	Clear hierarchy: Filled primary buttons vs. outlined secondary choices
Accessibility Integration	Robust: Detailed photo galleries document physical access features	Standard: Uses icons and text layouts across subpages
Content Strategy	Long numbered lists that require high scanning effort	Chunked horizontal content rows that prevent information overload
Microinteraction Style	Atmospheric visual effects (e.g., shooting stars)	Context-driven functional animations (e.g., orbital paths)

6. Overall Findings

Evaluating these reference platforms reveals several important trends for modern astronomical web systems. Balancing a site's commercial needs as a tourist attraction with its credibility as a research facility requires a structured layout strategy. Grouping layouts into distinct horizontal rows makes long pages easier to navigate and digest.

The analysis also highlights common challenges in transactional design. For instance, duplicating button labels for a single checkout action or using disjointed, off-site ticketing tools can create friction for users. Conversely, using targeted animations—like scroll-triggered reveals and context-driven motion—adds a layer of polish to complex informational pages.

7. Features to Adopt

- **Unified Thematic Framework:** Maintain a consistent, cohesive color palette across the entire application instead of altering background moods by section, ensuring strong system branding.
- **Task-Oriented Quick Menus:** Position a multi-column task navigation shortcut menu directly beneath the hero section to let different user types self-select their paths immediately.
- **Visual CTA Hierarchy:** Use filled button backgrounds for primary conversions (e.g., booking slots) and outlined treatments for secondary actions (e.g., documentation) to establish clear visual priority.

- **Context-Driven Motion:** Use scroll-triggered reveals (fade-in and slide-up) and content-matched vector motion (like orbit tracks) to add a polished feel to long text lists.
- **Segmented Archives:** Use tabbed navigation bars to split time-based programming into clean, manageable views (e.g., Current, Upcoming, Past).
- **Comprehensive Localization:** Ensure translation tools cover all site text, form controls, labels, and filter parameters evenly across the platform.
- **Visual Trust Markers:** Document physical accessibility options using real site photographs alongside descriptive text to help build user confidence.
- **Isolated Transaction Interfaces:** Strip out global navigation links during active checkout or booking processes to minimize distractions and help users complete forms.

8. Features to Avoid

- **Visual Discontinuity:** Avoid changing core color rules and styles between pages, which can weaken brand consistency and disrupt the user experience.
- **Overloaded Animation Layouts:** Avoid stacking high-resolution background assets alongside active layout animations to maintain fast page load times.
- **Duplicate CTA Mapping:** Do not map separate, uniquely labeled action buttons to the exact same checkout destination, which creates confusion.
- **Disjointed Booking Workflows:** Avoid forcing users into external ecosystems for ticketing or registration without offering an in-app path, keeping workflows smooth for international users.
- **Dense Text Lists:** Do not present policies, terms, or guidelines as long, unbroken numbered lists; use grouped layout blocks or collapsible accordions instead.
- **Decorative Hover Scaling:** Avoid applying hover-zoom transformations to gallery cards unless the enlargement reveals meaningful detail or data about the image.
- **Unnecessary Form Requirements:** Do not include non-essential data input fields in transactional checkouts, keeping friction low and completion rates steady.

9. Conclusion

Building a successful Planetarium & Observatory Management System requires a clear balance between informational content and streamlined transaction paths. The research highlights the value of using a single, consistent visual theme, clear call-to-action hierarchies, and well-structured, row-based layouts to keep the user experience smooth and engaging.

By incorporating smooth, scroll-triggered animations and native, distraction-free checkout paths while avoiding duplicate button actions and fractured external workflows, the upcoming system can deliver an efficient experience tailored to students, casual visitors, and researchers alike.

10. References

- <https://www.sstm-sam.org.cn/>

- <https://www.adlerplanetarium.org/>